

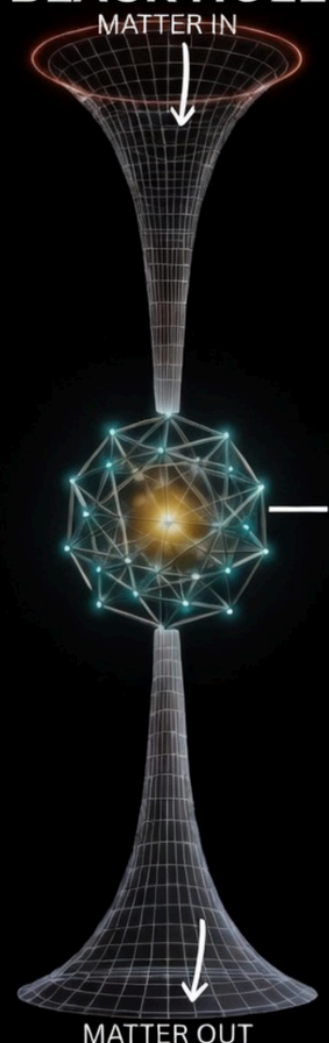


Newsletter

PIEAS Society for Physics

ZENITH

BLACK HOLE



Quantum Bounce
(Spin Networks)

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- Rotary Peace Fellowship
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Director's Note

Dear Readers,

It is my pleasure to welcome you to this edition of our newsletter. This platform serves as a reflection of the vibrant academic and research environment we are striving to cultivate, where curiosity, collaboration, and innovation come together to shape meaningful learning experiences.

Over the past months, our community has continued to explore new ideas and initiatives that contribute to scientific growth and intellectual exchange. From academic discussions and student-led activities to events, we have witnessed inspiring enthusiasm among students and researchers who are eager to engage with emerging fields and push the boundaries of knowledge.

This newsletter aims to highlight these efforts celebrating achievements, sharing opportunities, and documenting events that strengthen our academic culture. I hope it not only informs you about what we have accomplished, but also inspires greater participation and collaboration within our community. I would like to extend my sincere appreciation to everyone who contributed to making these activities possible. Your dedication and curiosity are what drive our community forward.

RUBAB RAHMAN

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INSPIRATION



Dr. Zaheer Asghar

Curiosity is something that has been directing me in the field of mathematics which is a language that reveals patterns in the nature and also solves complex questions in science. This love inevitably led me to become a mathematician and an academician, a profession that I have always wanted in life.

I graduated with my MSc and MPhil in Mathematics in Quaid-i-Azam University, Islamabad. I have started my PhD in Spring, 2012 in International Islamic University, Islamabad under the guidance of Prof. Dr. Nasir Ali. I joined Karachi Institute of Power Engineering (KINPOE) in December 2012 as a scientist during my doctoral studies. Balancing work and PhD research was not an easy task, and with the help of hard work, I managed to finish my PhD in approximately 5 years and published five research papers pertaining to qualitative analysis of fluid dynamics.

I have been teaching at different institutions since Fall, 2010, and one of that is PIEAS, where I taught different Mathematics courses at graduate and undergraduate levels.

INSPIRATION

In my opinion, the better-educated a teacher, the more powerful and stronger will be his or her students the knowledge and understanding rise in a result of successful mentoring.

My research interests span nonlinear analysis, fluid dynamics, machine learning, cryptology, quantum cryptography, and empirical modeling, and more recently in quantum computing and quantum machine learning. Heterogeneity of these regions indicates my attempt to match the mathematical study to contemporary and future scientific and engineering issues.

The most satisfying experience in my career has been to guide students in interdisciplinary research. I managed the students of the first batch of Chemical Engineering in the projects based on machine learning with Design of Experiments and the result was published in Bioresource Technology (impact factor 11.4 at the time). Other innovative concepts on which I have been working in the field of quantum cryptography are the application of chaotic systems to create randomness in a quantum key distribution. Recently, we have published our work on the aerodynamic design and optimization in Aerospace science and technology.

I would say to students that mathematics is a field that is fundamental to science and technological development. Linear algebra is the foundation of such areas as artificial intelligence and quantum computing, whereas abstract algebra and number theory are at the heart of post-quantum cryptography, which will protect cyberspace against future quantum attacks. The need to have such foundations is highlighted in initiatives made by bodies like NIST. I want my students to be curious, work hard, and appreciate mathematics because what may come out as an abstract concept in this day and age may turn out to be the breakthroughs in the near future.

STUDENTS' PROJECT

BLACK HOLES: A QUANTUM GRAVITY PERSPECTIVE

AUTHORED BY: SYED ABBAS AHMAD — MOHAMMAD ABDULLAH JAVAID (BS PHYSICS 21-25)
SUPERVISED BY: DR. KHALID SAIFULLAH AND DR. NADEEM SHAUKAT

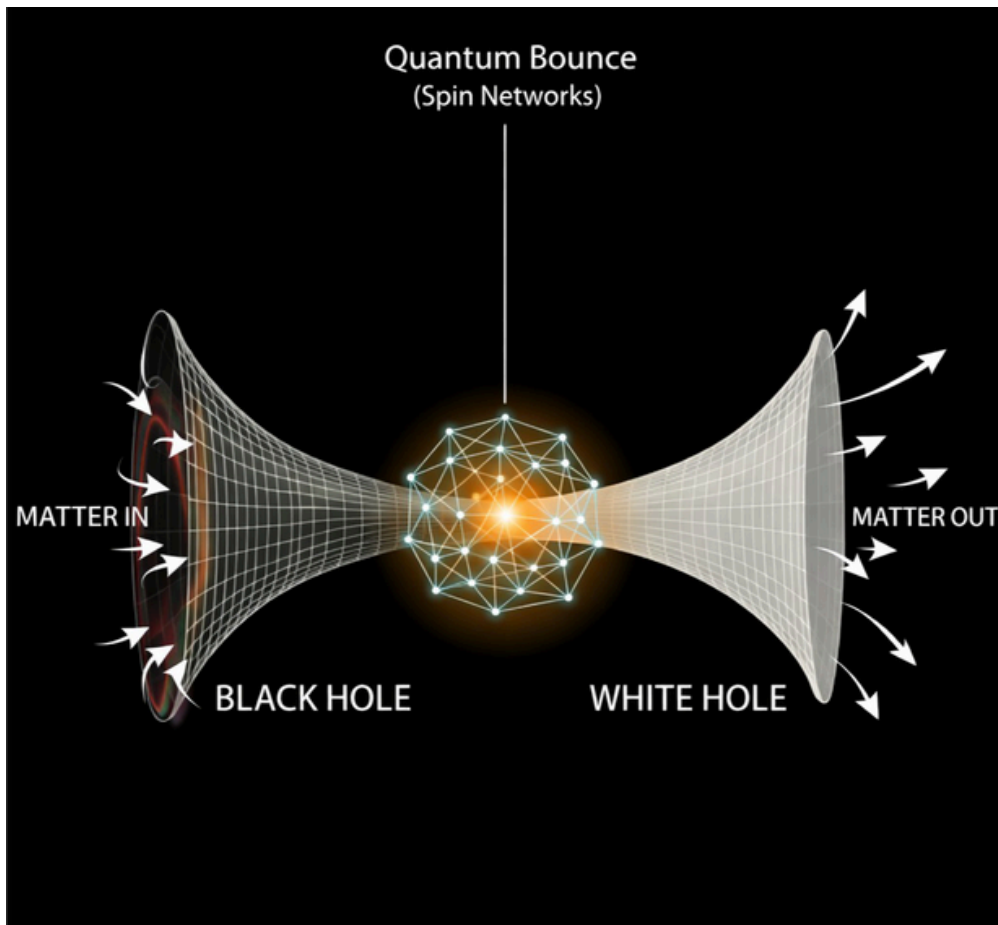
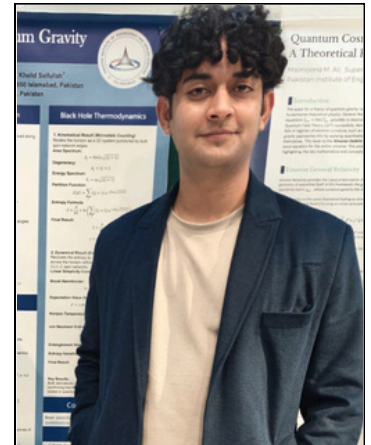


FIGURE 1: A SCHEMATIC OF A QUANTUM BOUNCE VIA SPIN NETWORKS, SHOWING MATTER ENTERING A BLACK HOLE AND EXITING A WHITE HOLE IN A LOOP QUANTUM GRAVITY FRAMEWORK.



Syed Abbas Ahmad



Mohammad Abdullah Javaid

General Relativity explains gravity as a manifestation of spacetime curvature, with the gravitational field encoded in the geometric structure of spacetime rather than existing as an independent entity identical to it. Within this framework black holes emerge as the most extreme gravitational objects, representing regions where gravity becomes so strong that cannot escape. However, General Relativity is a classical theory that treats spacetime as smooth and continuous, and this description fails under the extreme conditions inside a black hole.

Two major issues arise: the singularity problem and the entropy problem. Gravitational collapse leads to a singularity of infinite density, signaling the breakdown of classical physics. Meanwhile, black hole

BLACK HOLES: A QUANTUM GRAVITY PERSPECTIVE

thermodynamics introduces the Bekenstein–Hawking entropy, proportional to the horizon area in Planck units. Since entropy is fundamentally microscopic, a purely classical black hole—defined only by macroscopic parameters like mass and charge—lacks the degrees of freedom required to explain it. This motivates a quantum description.

Loop Quantum Gravity (LQG) provides such a framework by introducing a fundamental length scale, the Planck length ($\sim 10^{-33}$ cm), beyond which distance loses operational meaning. Probing smaller scales would require energies sufficient to form a black hole, implying spacetime is not infinitely divisible. The Planck scale thus acts as a natural cutoff, preventing classical divergences.

In LQG, spacetime is fundamentally discrete. Geometric operators such as area and volume possess discrete spectra, analogous to quantized energy levels in quantum mechanics. The fundamental excitations of geometry are organized into spin networks composed of nodes and links. Nodes represent elementary volumes, links represent elementary areas, and their quantum labels determine precise geometric properties. The network structure

encodes how space is assembled from these basic units.

Applied to black holes, this framework yields a statistical derivation of Bekenstein–Hawking entropy.

The horizon is modeled as punctured by spin network links, each contributing a quantized area. Fixing the macroscopic area, one counts all quantum configurations compatible with it; the logarithm of this count gives the entropy. This reproduces the area law with small logarithmic corrections, providing a microscopic foundation for black hole thermodynamics.

LQG also addresses the singularity problem. Because spacetime cannot be compressed below the Planck scale, collapse does not produce infinite density. Instead, it reaches a minimum scale and undergoes a quantum bounce. Effective models suggest the black hole interior transitions into an expanding region resembling a white hole. In this way, a consistent quantum theory of gravity resolves classical singularities and offers a coherent description of extreme gravitational regimes.

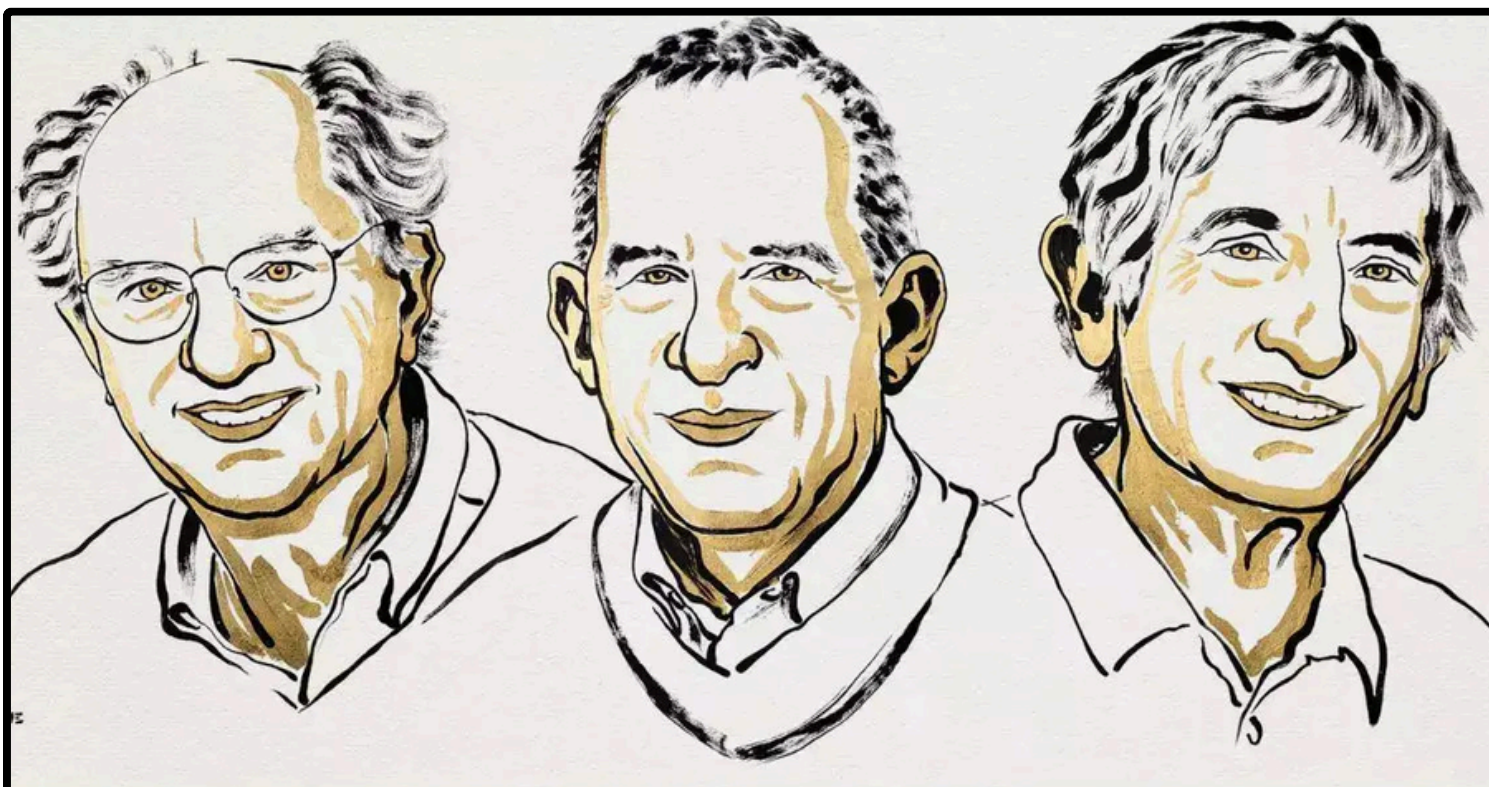
nobel

2025

prize

The 2025 Nobel Prize in Physics was awarded to **John Clarke**, **Michel H. Devoret**, and **John M. Martinis** for demonstrating macroscopic quantum

tunnelling and energy quantization in superconducting circuits. Using Josephson junction devices, they showed that an electrical circuit can behave like a single quantum system. Their experiments revealed quantum tunnelling through an energy barrier and discrete energy absorption, confirming quantum behavior at a macroscopic scale.



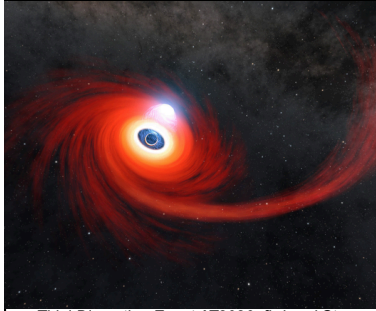
John
Clarke

Michel H.
Devoret

John M.
Martinis

"for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation"

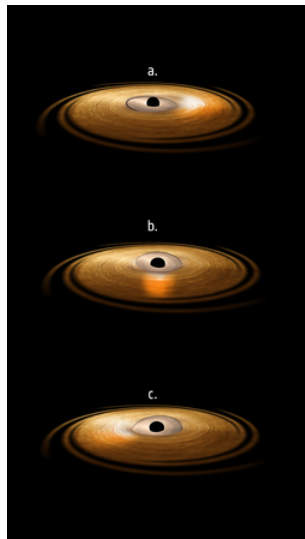
Physics Bulletin



Tidal Disruption Event AT2020afhd and Star Destruction



Observational Methods and Multi-Wavelength Approach



a feasting black hole

Frame-Dragging Confirmed Near a Black Hole

In 2025, astronomers confirmed frame-dragging, a concept from Einstein's General Theory of Relativity, during the observation of the tidal disruption event AT2020afhd, where a star was destroyed by a spinning supermassive black hole. The black hole's rotation caused spacetime distortion, leading to a wobble in the accretion disk and relativistic jet, analyzed using data from NASA's Swift Observatory and the Karl G. Jansky Very Large Array. Led by researchers from the National Astronomical Observatories of the Chinese Academy of Sciences, with contributions from Dr. Cosimo Inarra and international teams, the study provided direct evidence of spacetime being affected by a black hole's rotation, supporting relativity's predictions in extreme gravity.

Sources: Nature Astronomy; Phys.org; NASA Swift Mission releases

Longest Gamma-Ray Burst Ever Detected

In 2025, astronomers recorded the longest gamma-ray burst, lasting over seven hours, far longer than typical bursts. Detected by NASA's Fermi Gamma-ray Space Telescope and followed by other observatories, the event was analyzed by teams from Carnegie Mellon University and others. The burst's unusual duration suggests a new physical mechanism, potentially involving prolonged accretion or a rare black hole-star interaction, challenging existing gamma-ray burst models and indicating a new class of high-energy cosmic explosions.

Sources: The Astrophysical Journal Letters; Space.com; NASA Fermi Mission



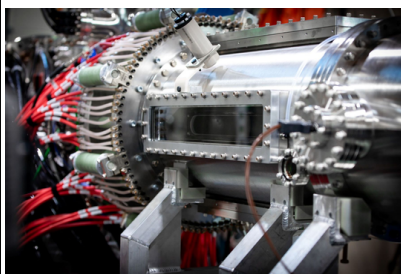
Artistic Representation of GRB 250702B

GRB 250702B

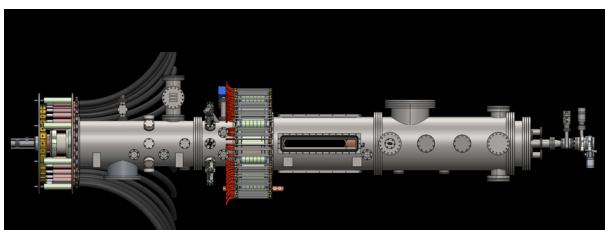


GRB 250702B (circled in red), taken on 3 July with the VLT's HAWK-I infrared camera

Event type	Gamma ray burst
Right ascension	18 ^h 58 ^m 45.565 ^s
Declination	-7° 52' 26.42"
Redshift	≤ 1



Operating a new device named the Fusion Z-pinch Experiment 3, or FuZE-3, Zap Energy has now achieved plasmas with electron pressures as high as 830 megapascals (MPa), or 1.6 gigapascals (GPa) total, comparable to the pressures found deep below Earth's



Compact Fusion Device Reaches Stellar Pressures

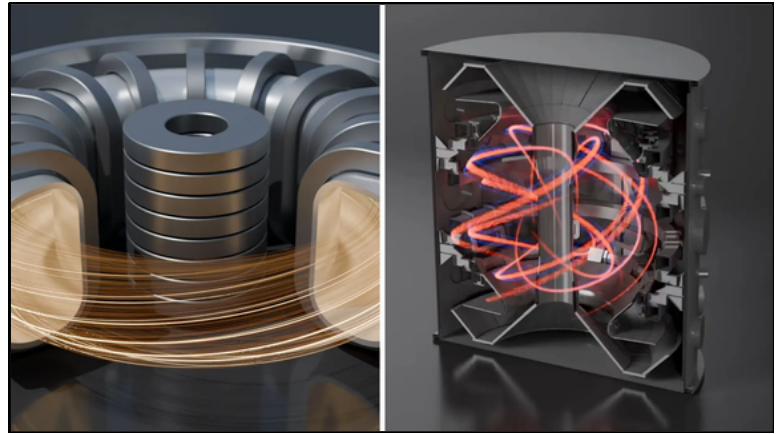
In 2025, Zap Energy Inc. achieved gigapascal-level plasma pressures in its compact fusion device, FuZE-3, using a sheared-flow-stabilized Z-pinch configuration. This milestone showed that extreme fusion conditions can be reached without large tokamak reactors, enhancing understanding of high-energy-density plasma physics and supporting smaller, simpler, and potentially more economical fusion energy approaches.

Sources: EurekAlert!; SciTechDaily; Zap Energy technical releases

Physics Bulletin

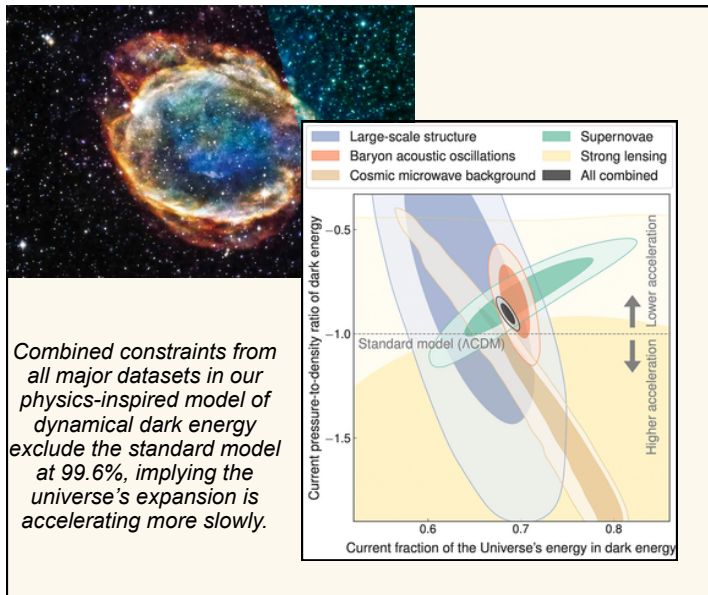
AI Accelerates Nuclear Fusion Research

Scientists at the UK Atomic Energy Authority (UKAEA) advanced fusion research by using artificial intelligence to model plasma turbulence, a key challenge in controlled nuclear fusion. AI trained on experimental data predicted plasma behavior much faster than traditional methods. This breakthrough allows for quick testing of reactor designs and control strategies, speeding up the development of practical fusion energy and showcasing AI's potential to overcome computational barriers in experimental physics.



A fusion plasma simulation embedded into UKAEA's MAST Upgrade machine.

Sources: UKAEA reports; *Nature Physics*; *ScienceDaily*



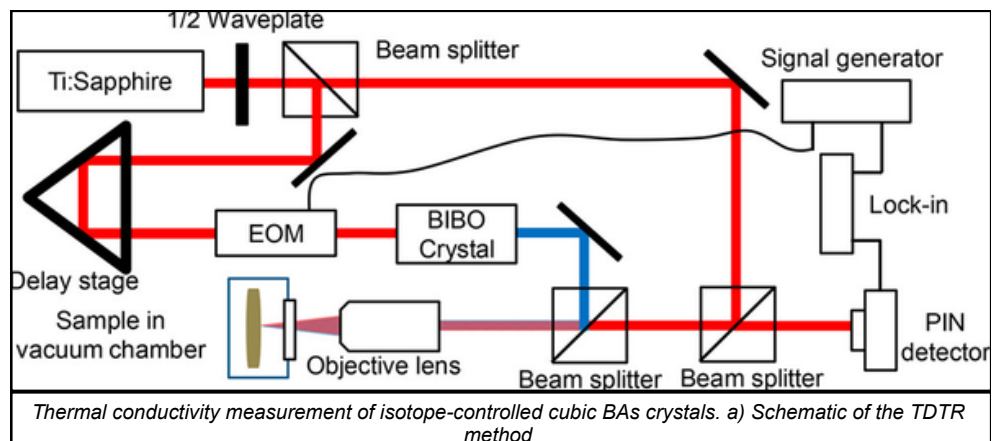
Dark Energy May Be Changing Over Time

In 2025, scientists from collaborations like Fermilab and the University of Chicago found evidence that dark energy might change over time, challenging the standard cosmological model that assumes it is constant. By analyzing galaxy distributions, they detected deviations from predictions based on a constant cosmological constant. These provisional findings suggest that evolving physics could drive the universe's accelerated expansion, potentially requiring significant revisions in modern cosmology if confirmed.

Sources: *Physical Review D*; Fermilab publications; *Nature Astronomy*

Boron Arsenide Surpasses Diamond in Heat Conduction

Materials scientists from the U.S. and China found that boron arsenide conducts heat more efficiently than diamond under controlled conditions. Led by Professor Zhifeng Ren, the research showed that improved crystal purity significantly enhances thermal transport. This confirms a long-standing theory and positions boron arsenide as a top thermal conductor, with important implications for thermal management in high-performance electronics, power devices, and quantum technologies.



Sources: *Science*; *ScienceDaily*; University of Houston press releases

ALUMNI SPOTLIGHT



HARRON AHMED

Haroon Ahmed developed and implemented a balanced homodyne detection system for squeezed light measurement as his Final Year Project. A technique essential for high precision quantum optical sensing. The project involved designing a complete optical setup with beam splitters, coherent laser sources, and matched photodiodes, along with Arduino based data acquisition system, with a particular focus on signal conditioning and noise reduction to accurately measure quantum noise. He is now an Erasmus Mundus Scholar under the Condensed Matter Photonics Joint Master's programme, with a monthly stipend of approximately €1,400 and full tuition coverage, studying at the University of Eastern Finland with mobility across institutions in France, Finland, and Lithuania. His advice: Stay curious, build strong foundations, seek hands-on experience, apply boldly for international scholarships and above all, be patient and persistent, because dedication today opens doors you cannot yet imagine.

HIRA JAVED

Hira Javaid completed her Bachelor's FYP in Experimental Condensed Matter Physics, where she synthesized pure and Manganese doped Copper Sulfide using the solid state reaction method and characterized the materials via XRD and electrical IV measurements to study their structural and electrical properties. She believes the specific FYP topic matters less than the skills and curiosity built along the way, a philosophy that has clearly served her well. Hira is now an Erasmus Mundus Scholar in the QuanTEEM (Quantum Technologies and Engineering) Master's programme, currently studying at Université de Bourgogne in France, with the scholarship covering full tuition and offering mobility across Europe's leading research institutions. She advises juniors applying to European programmes that CGPA is only one piece of the puzzle: what truly matters is a strong, relevant CV backed by practical skills in tools like MATLAB and Python, and the ability to present your experience with clarity and confidence.



MUHAMMAD SOHAIL AHMED

Muhammad Sohail Ahmad (Batch 20-24) explored the phenomenon of atomic coherence through his FYP titled "Study of Atomic Coherence in Electromagnetically Induced Grating," in which he compared the degree of polarization rotation of a probe beam in EIT and EIG configurations, research with direct applications in laser frequency stabilization, laser locking, and quantum sensing. He is now pursuing a Master's degree in Optics at Shanxi University, Taiyuan, China, where his research focuses on continuous variable quantum information processing, including a current project on Bessel beams as candidates for quantum dense coding. His scholarship from Shanxi University covers full tuition, accommodation, and a monthly stipend, and he studies within the Institute of Optoelectronics, which is ranked 2nd in China for quantum research. His message to juniors: Dream beyond borders, believe beyond limits, stay curious, work hard, and remember your future is bigger than you think.



ALUMNI SPOTLIGHT

MUSTAFA SHAH

Mustafa Shah (Batch 20–24) conducted his Final Year Project on superconductivity in 2D materials, contributing to one of the most actively studied frontiers in condensed matter physics. He is currently affiliated with the State Key Laboratory of Quantum Optics and Quantum Optical Devices at Shanxi University, Taiyuan, China a world class research environment where he continues to deepen his expertise. Mustafa shares his wisdom with juniors in characteristically grounded terms: keep going no matter what, stay humble, work as if everything depends on you and pray as if it all depends on God. He also adds a touch of humor that he was never particularly good at physics, and arguably still isn't, but the key difference now is that he gets paid for it rather than punished.



KIRAN FATIMA KHIYANI

Kiran Fatima Kayani (Batch 21–25) investigated quantum entanglement in optomechanical systems for her FYP, analyzing both bipartite and tripartite configurations to study the generation, stability, and robustness of quantum correlations under realistic decoherence conditions, research that contributes to the theoretical foundations of hybrid quantum systems for quantum communication, sensing, and information processing. She has been awarded a fully funded Direct PhD scholarship at King Fahd University of Petroleum and Minerals (KFUPM) in Saudi Arabia, currently ranked 67th globally and 1st in the Middle East, with close ties to Aramco. The scholarship is exceptionally comprehensive, covering free housing, an annual round trip air ticket, subsidized meals, full medical coverage, and conference funding. Kiran encourages juniors to stay curious, consistent, and confident, reminding them that success is not achieved overnight it comes through patience, dedication, and the courage to dream big.



SYED HAIDER ALI

Syed Haider Ali (Batch 21–25) completed his Final Year Project on the multivariate analysis of four top quark production in association with a W boson, employing advanced deep learning techniques including Convolutional Neural Networks (CNNs) and Quantum Graph Neural Networks (QGNNs). His work sits at the cutting edge of high energy physics and machine learning, reflecting strong research ambition from the very start of his career. He has since been awarded the prestigious Stipendium Hungaricum Scholarship, a fully funded opportunity to pursue graduate studies at the University of Debrecen, Hungary. His advice to juniors is refreshingly straightforward: apply for as many programs as you can, because opportunity favours those who try.



PSP EVENTS



QUANTUM COMPUTING HACKATHON 2026

The first national Quantum Computing Hackathon was held from 6–8 February 2026 at the National Centre for Physics, Islamabad. The event



brought together students and researchers to develop quantum-based solutions addressing real-world challenges linked to the UN Sustainable Development Goals. The hackathon was organized by Pakistan Institute of Engineering and Applied Sciences and National Institute of Lasers and Optronics under the Pakistan Atomic Energy Commission, with

support from the Open Quantum Institute and CERN. More than 950 applications were received nationwide, with 42 finalists forming seven teams for the final round. Participants worked with mentors and presented innovative quantum computing ideas, highlighting the growing momentum of quantum research in Pakistan.

CAN WE REVERSE DECOHERENCE ?



On December 16, 2025, CMS and PSP welcomed Prof. Muhammad Suhail Zubairy of Texas A&M University for an insightful lecture titled “Can We Reverse Decoherence?”. He discussed how weak measurements can both trigger and potentially reverse decoherence, a key challenge in quantum science. He highlighted the significance of this work for quantum computing and future quantum technologies, sparking thoughtful discussion among students.



MATHLETICS 2025

From December 8–11, 2025, the PIEAS Society for Physics (PSP), in collaboration with ASME, hosted Mathletics 2025, challenging students across PIEAS with written and on-board rounds in integration, differentiation, trigonometry, and geometry.

After four days of intense problem-solving, the closing ceremony celebrated the winners:

Hooriya Zahid, Saniya Rahim Abbasi, and Zain Ali, who received cash prizes and shields.

All participants and the organizing team were recognized with certificates for their dedication. The event highlighted analytical skills, perseverance, and a passion for mathematics, inspiring the next generation of problem-solvers.

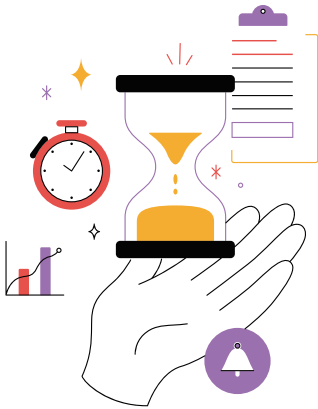
Pinboard

1 Australia Awards Scholarships

Host: Australian Government

Fully funded scholarships for undergraduate, master's, and PhD programs in Australia.

Deadline: 30 April 2026



4 Rotary Peace Fellowship

Host: Rotary International

Fully funded master's program for students interested in peace and conflict studies.

Deadline: 15 April 2026.



2 Think Big Scholarship

Host: University of Bristol

Scholarships for international undergraduate and postgraduate students in the UK.

Deadline: 29 April 2026.

5 UCL Global Undergraduate Scholarship

Host: University College London

Financial support for international undergraduate students studying in the UK.

Deadline: 27 April 2026

3 Joint Japan/World Bank Graduate Scholarship

Host: World Bank & Government of Japan

Master's scholarships for students from developing countries.

Deadline: 27 April 2026 (first round)



6 Orange Knowledge Programme Scholarship

Host: Government of the Netherlands

Scholarships for short courses and master's programs in the Netherlands.

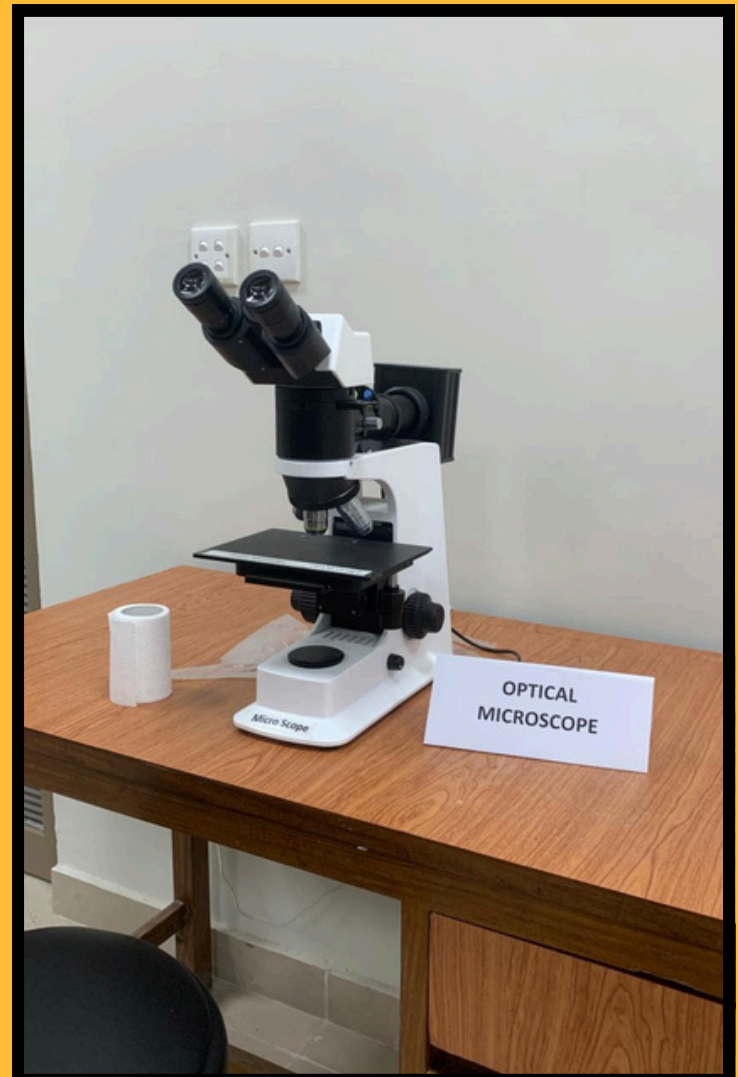
Deadline: 22 April 2026 (varies by course)

CMS LABORATORIES

CENTRAL FOR MATHEMATICAL SCIENCES

CMS is a research centre at Pakistan Institute of Engineering & Applied Sciences (PIEAS) focused on advancing core and applied mathematics, fostering research collaborations, hosting workshops and conferences, and nurturing young mathematicians with world-class facilities and visiting experts.

Device Characterization Lab



Nanostructure Synthesis Lab



Physics Humour Corner

People act like Isaac Newton was so smart,
but he didn't even learn Calculus until he was 30.

According to the Second Law of Thermodynamics, entropy always increases.
So technically...
your room getting messier over time is not laziness,
it's physics.

A farmer asks a physicist to help increase milk production.
The physicist returns after a week and says:
"I have a solution...
but it only works for a spherical cow in vacuum."

In physics labs:
If the experiment works → publish it.
If it doesn't work → adjust the error bars.

First week of Quantum Mechanics:
Student: "I don't understand this."
Professor: "That's normal."
Mid-semester:
Student: "I think I understand it."
Professor: "That's concerning."

In physics papers:
Small error bars → "We measured very carefully."
Large error bars → "We measured... very honestly."

How many theoretical physicists does it take to change a light bulb?
Two.
One to hold the bulb
One to rotate the universe

During physics exams:
If the answer looks too simple → it's wrong.
If the answer looks terrifying → it's probably wrong too.